26900227 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Core Course (DSC) - MG5DSCFIM300 - GYM INSTRUCTOR ESSENTIALS

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Fitness programmes for older adults should primarily emphasize: [U] / [CO1, CO5]

   a). Functional capacity b). Maximum training loads

   c). Competitive performance d). Sport specialization

2. Heart rate during gym-based exercise can be monitored using: [U] / [CO2, CO4]

   a). Heart-rate monitor b). Stadiometer

   c). Measuring tape d). Goniometer

3. A programme includes 3 sets of squats, 3 sets of lunges, and 2 sets of leg curls. What is the total number of sets? [A] / [CO1]

   a). 6 sets b). 7 sets

   c). 8 sets d). 9 sets

4. Regular physical activity contributes most directly to: [U] / [CO1]

a). Improved health

b). Increased inactivity

c). Reduced mobility

d). Lower work capacity

5. Internal load is primarily reflected by the body's response to: [U] / [CO1, CO2]

a). Training stimulus b). Exercise equipment

c). Training venue d). Programme duration

6. A trainer increases the training frequency from three to four sessions per week. Which load factor has changed? [An] / [CO1]

a). Volume b). Intensity c). Frequency d). Duration

7. Which is an important emergency safety measure in a gym [K] / [CO2]

a). Keeping emergency exits blocked b). Knowing the location of first-aid equipment and emergency exits

c). Storing equipment near exits d). Turning off all ventilation

8. The purpose of cardiopulmonary resuscitation (CPR) is primarily to maintain: [U] / [CO1]

a). Joint mobility

b). Muscle flexibility

c). Circulation and oxygenation

d). Body temperature

9. Which document is useful for organizing participants and recording their test results [U] / [CO2, CO4]

a). Attendance/testing schedule sheet b). Newspaper

c). Advertisement d). Certificate only

10. Which measurement is used to assess blood pressure [K] / [CO2, CO4]

a). Stadiometer b). Sphygmomanometer

c). Goniometer d). Spirometer

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Why is hydration important throughout life? [U] / [CO2, CO4]

12. Explain the importance of coordination in performing skilled movements. [U] / [CO1]

13. What are micronutrients? Give examples. [U] / [CO2, CO4]

14. How does physical fitness influence performance of daily activities? [U] / [CO1]

15. How can a fitness trainer improve client motivation? [U] / [CO2]

16. How can a gym instructor apply overload when resistance cannot be increased safely? [A] / [CO1]

17. What is hydration? Why is it important during exercise? [U] / [CO2, CO4]

18. What is progressive loading? [K] / [CO1]

19. What is a macrocycle? [K] / [CO2, CO4]

20. A conscious casualty is frightened and confused after an injury. How should the first aider respond? [A] / [CO1]

21. What is meant by total training load? [U] / [CO1]

22. Define flexibility as a health-related fitness component. [K] / [CO1]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

23. Explain the importance of recovery capacity when applying progressive overload. [U] / [CO1]

24. A trainee sustains an acute injury during exercise. Explain how the gym instructor should respond while ensuring the safety of the casualty and others present. [An] / [CO1, CO2]

25. Explain iodine, its functions, sources, and deficiency disorders. [U] / [CO2, CO4]

26. Explain why the same repetition total may be distributed across sets in different ways and how this can affect training demand. [An] / [CO1]

27. What is progression? Explain how it should be applied when planning an exercise programme. [A] / [CO2, CO5]

28. Explain the functions of fats in the human body and differentiate between saturated and unsaturated fats. [U] / [CO2, CO4]

[5x4 = 20]